

SuperFlow: Invoice Reconciliation

LESSON 04 · DESIGN & CREATE

What you'll learn

- Read a real SuperFlow end to end — trigger, document parsing, extraction, a branch, a human approval, and two exits
- Mix deterministic logic and agentic judgment on one canvas, sometimes in the very same node
- Pause a run durably for human approval — for minutes, hours, or days — at no cost
- Replay and resume runs from history without re-running steps that already completed

Key concepts

SuperFlow is Lyzr's visual builder for systems of agents — deterministic where you need guarantees, agentic where you need judgment. Hand work to a manager and you get intelligence but lose control over *when* things happen; wire a rigid static workflow and you get control but hit a ceiling fast. SuperFlow refuses that trade-off: you choose, node by node, which cells are guaranteed and which are judged.

Every SuperFlow starts with a trigger, and one trigger can start three ways. The invoice flow's trigger accepts **file uploads**; the same trigger also exposes a **webhook URL with a secret** so another system can fire it programmatically, and a **cron schedule** to run it on a timetable. One node, three ways to start.

Document AI nodes turn a raw file into structured fields. **Parse document** takes whatever the trigger received — PDF, Word doc, image, same node — and extracts its text. **Extract fields** then runs an extraction model against that text using a schema you define; here it returns an invoice ID and a total amount as structured fields you can reference downstream.

The same node can be deterministic or AI-judged — that's the core idea. **Match SOR** is an **if** node checking whether the extracted invoice ID equals an ID in your system of record: match routes one way, no-match the other. But the same **if** has an **AI mode** — for a fuzzy condition like "does this invoice look unusual," you write it in plain language and let an LLM evaluate it. Some cells you want guaranteed; some you want judged.

A **wait for approval** node pauses the run durably — and the pause is free. On the no-match branch, an **approval-summarizer agent** writes a short paragraph for the human, who then sees a form with that summary

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and a field to paste a corrected invoice. The run holds there for minutes, hours, or days at no cost; approve and it resumes, reject and it ends with an error.

Replayable history is what makes it production-grade, not a toy canvas. Every run is saved; you can open history, replay any execution on the canvas, and rerun from a specific node. Steps that already completed don't run again — so a mid-flow retry won't double-charge anyone, approvals can wait days, and schedules survive restarts.

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In Studio

Studio feature: **Create Agent → SuperFlow** (the visual flow builder, document-AI nodes, **if** node, wait-for-approval, HTTP request nodes).

Walking the **invoice reconciliation** flow node by node:

1. **Trigger** — accepts a file upload (an invoice). Also offers a webhook URL + secret and a weekday cron, if another system or a schedule should start it.
2. **Parse document** — a document-AI node; extracts text from the uploaded PDF / Word / image.
3. **Extract fields** — runs an extraction model with your schema; pulls the **invoice ID** and **total amount** as structured fields.
4. **Match SOR** — an **if** node comparing the extracted ID against your system of record. (*In production this would be an HTTP request to your database just before the **if**; the demo hard-codes one ID to keep the flow in focus.*)
5. **Match → S3**. On a match, an **HTTP request** node posts the original file straight to S3 — no human in the loop.
6. **No match → approval summarizer**. An agent (built earlier in the agents tab) reads the extracted data and writes a short summary for the approver.
7. **Wait for approval** — the run pauses. The approver sees the summary plus a field to paste the corrected invoice's URL. Durable pause; approve to continue, reject to end with an error.
8. **Recovery → S3**. On approval, an HTTP request posts the **corrected** file to S3.
9. **Run it with a non-matching invoice** and watch each node light up; the false branch summarizes, pauses (blue = waiting), then resumes on approval and posts the corrected file. Open **history** to replay or rerun from any node.

TRY THIS

- 1 Trace a flow you care about as a SuperFlow on paper: name the trigger, the parse/extract steps, the one **if** that branches it, and the two terminal exits. Mark which single node would need a human approval.
- 2 For your branching **if**, decide whether it should be **deterministic** (an exact match) or **AI-judged** (a fuzzy "does this look off?"). Write the condition both ways and notice which one your case actually needs.
- 3 Add a **wait-for-approval** step in your design and write the summary an agent should hand the approver. *Keep it to what they need to decide* — what it is, the amount, and what's worth flagging.